Bivariate Descriptive Statistics & Probability Distributions

Lecture 3 - LAS

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07-09-2026

Today

- Bivariate Descriptive Statistics
- Transformations
- Bivariate Distributions

Today

Chapter 5 and Chapter 6 from the gitbook.

Bivariate Descriptive Statistics

What is “Bivariate”?

Looking at two variables at the same time.

Yes, there is also “trivariate” and more generally, “multivariate”.

Bivariate Descriptive Statistics

- Contingency tables (we talked about these last week)
- Covariance
- Correlation

Dataset: World Data

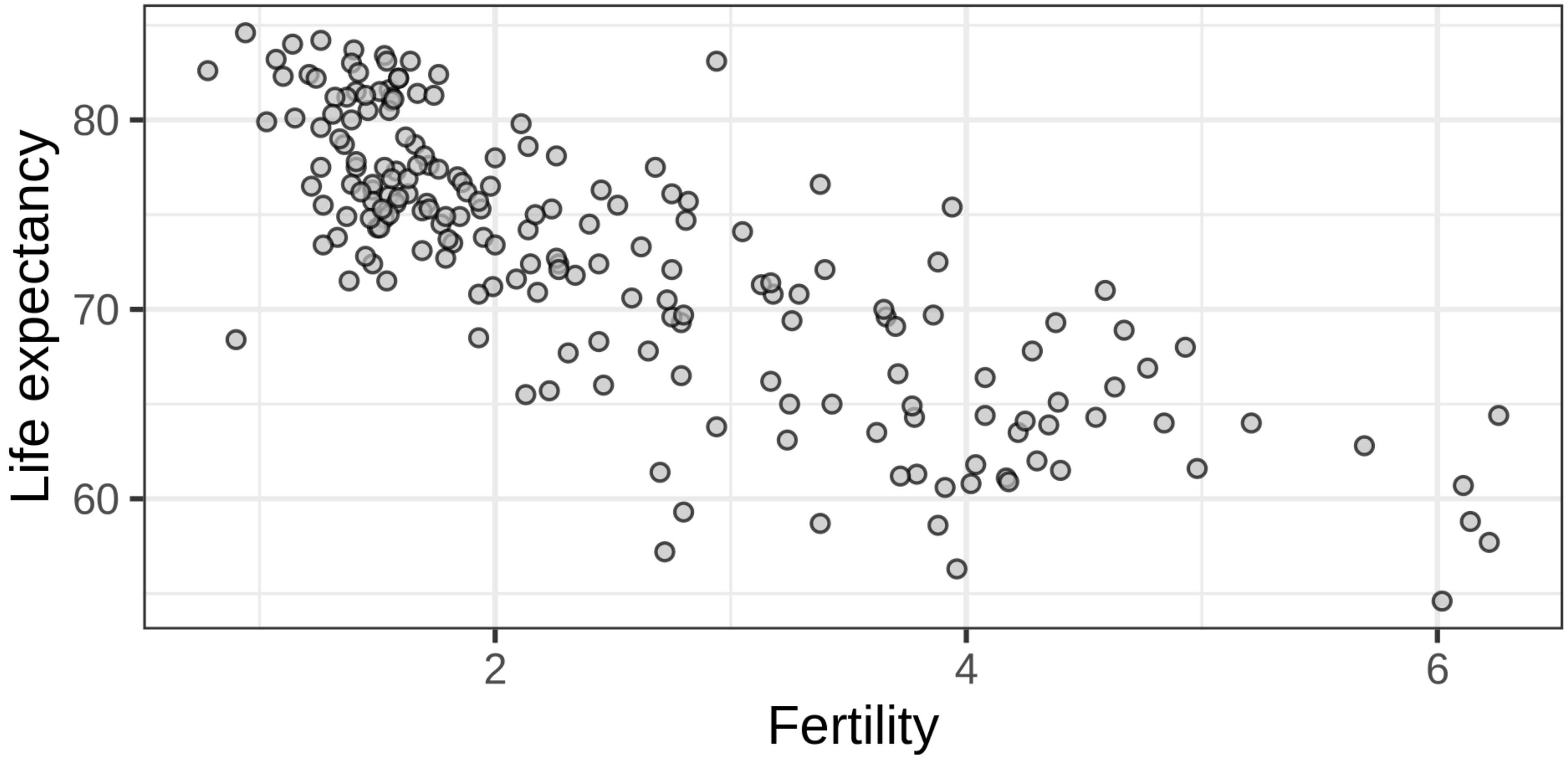
NAME	CHILD_MORTALITY	FERTILITY	CO2_PCAP	GDP_PCAP	LIFE_EXP
Afghanistan	52.52	4.93	0.28	1,981.71	
Angola	69.33	5.21	1.23	7,397.49	
Albania	9.06	1.36	2.10	17,111.95	
Austria	2.91	1.41	9.20	65,661.45	
Belgium	3.47	1.55	16.93	62,744.20	
Brazil	13.78	1.63	2.11	18,554.05	

Source: Data from world bank via https://www.gapminder.org/data, CC-BY license

Dataset: World Data

- Child mortality: Probability of dying between age 0 - 5 per 1000 babies (percentage).
- Fertility: This is the number of children that would be born to a woman if she were to live to the end of her childbearing years and bear children in accordance with age-specific fertility rates of the specified year.
- CO2_PCAP: CO2 emissions per capita.
- GDP_PCAP: GDP per capita.
- Life Expectancy: No. years a newborn would live at current mortality rates.
- Daily income: The mean daily household per capita income in 2021 PPP (constant international dollars).

Fertility vs. Life expectancy



Last Week: Sum of Squared Distances SS_x

$$SS_x = \sum_{i=1}^{n} (x_i - \bar{x})^2 = \sum_{i=1}^{n} (x_i - \bar{x})(x_i - \bar{x})$$

One variable: $\text{deviation} \times \text{deviation} = \text{Variance}$

Two variables: $x\text{—deviation} \times y\text{—deviation} = \text{Covariance}$

Covariance: Sum of Products

$$SP_{xy} = \sum_{i=1}^{n} \left(x_i - \bar{x}\right)\left(y_i - \bar{y}\right)$$

Definitions:

- x and y: the two variables we're interested in.
- $\bar{x}$: the sample mean of the variable x.
- x_i: the ith observation of the variable.
- n the sample size.
- $\sum_{i=1}^{n}$: the sum from i is 1 to i is n.
- SP_{xy} the sum of products between x and y.

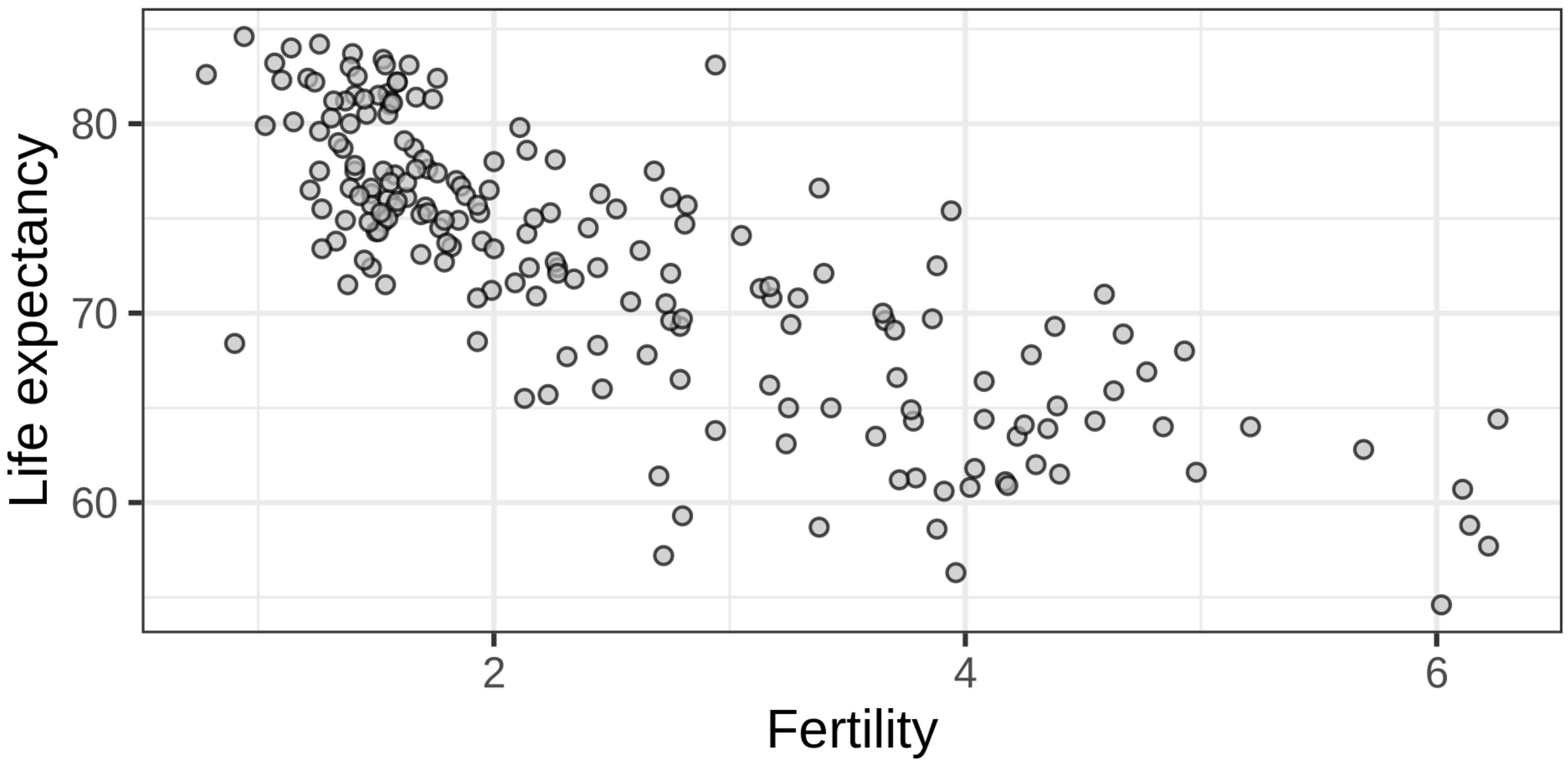
Covariance: Sum of Products

$\overline{\text{Life Expectancy}} = 72.8$, $\overline{\text{Fertility}} = 2.5$,

NAME	LIFE_EXPECTANCY	FERTILITY	LIFE_EXPECTANCY_M0	FERTILITY_
Afghanistan	68.00	4.93	-4.75	
Angola	64.00	5.21	-8.75	
Albania	78.70	1.36	5.95	
Austria	81.50	1.41	8.75	
Belgium	81.60	1.55	8.85	
Brazil	76.10	1.63	3.35	

Covariance: Sum of Products

SP_{xy} = -1326. What does this mean?



Covariance

$$S_{xy} = \frac{\sum_{i=1}^{n}\left(x_i - \bar{x}\right)\left(y_i - \bar{y}\right)}{n} = \frac{SP_{xy}}{n}$$

Definitions:

- x and y: the two variables we're interested in.
- $\bar{x}$: the sample mean of the variable x.
- x_i: the ith observation of the variable.
- n the sample size.
- $\sum_{i=1}^{n}$: the sum from i is 1 to i is n.
- SP_{xy} the sum of products between x and y.

S_{xy} = -6.84.

Correlation: Standardized Covariance

$$r_{xy} = \frac{\sum_{i=1}^{n} \left(x_i - \bar{x}\right)\left(y_i - \bar{y}\right)}{n s_x s_y} = \frac{SP_{xy}}{n s_x s_y} = \frac{S_{xy}}{s_x s_y}$$

Definitions:

r_{xy} = -0.791.

- x and y: the two variables we're interested in.
- $\bar{x}$: the sample mean of the variable x.
- x_i: the ith observation of the variable.
- n the sample size.
- $\sum_{i=1}^{n}$: the sum from i is 1 to i is n.
- SP_{xy}: the sum of products between x and y.
- s_x: the sample standard deviation of x

Correlation: Standardized Covariance

Recall Z-scores, $Z_{x_i} = \frac{x_i - \bar{x}}{s_x}$ and $Z_{y_i} = \frac{y_i - \bar{y}}{s_y}$

$$r_{xy} = \frac{\sum_{i=1}^{n} \left(x_i - \bar{x}\right)\left(y_i - \bar{y}\right)}{n s_x s_y}$$

$$r_{xy} = \frac{1}{n} \sum_{i=1}^{n} \frac{\left(x_i - \bar{x}\right)}{s_x} \frac{\left(y_i - \bar{y}\right)}{s_y}$$

$$r_{xy} = \frac{1}{n} \sum_{i=1}^{n} \underbrace{\frac{\left(x_i - \bar{x}\right)}{s_x}}_{Z_{x_i}} \underbrace{\frac{\left(y_i - \bar{y}\right)}{s_y}}_{Z_{y_i}}$$

$$r_{xy} = \frac{1}{n} \sum_{i=1}^{n} Z_{x_i} Z_{y_i}$$

Covariance Matrix

	CHILD_MORTALITY	FERTILITY	CO2_PCAP	GDP_PCAP	LIF
child_mortality	663.67	27.33	-68.93	-423,932.21	
fertility	27.33	1.56	-3.35	-19,479.86	
co2_pcap	-68.93	-3.35	28.41	107,317.28	
gdp_pcap	-423,932.21	-19,479.86	107,317.28	1,553,199,101.54	
life_expectancy	-149.48	-6.87	20.76	141,599.05	
daily_income	-445.96	-20.18	105.65	2,029,933.87	

Correlation Matrix

	CHILD_MORTALITY	FERTILITY	CO2_PCAP	GDP_PCAP	LIFE_E
child_mortality	1.00	0.85	-0.50	-0.42	
fertility	0.85	1.00	-0.50	-0.40	
co2_pcap	-0.50	-0.50	1.00	0.79	
gdp_pcap	-0.42	-0.40	0.79	1.00	
life_expectancy	-0.83	-0.79	0.56	0.52	
daily_income	-0.32	-0.30	0.78	0.96	

Covariance vs. Correlation

Covariance

- $-\infty < S_{xy} < \infty$
- unit, $\mathrm{unit}(x) \times \mathrm{unit}(y)$
- covariance of x with x equals variance
- scales with linear transformation of the data (e.g., mm to cm)
- cannot be interpreted without knowing the variances
  - but the sign tells us if the relation is positive or negative

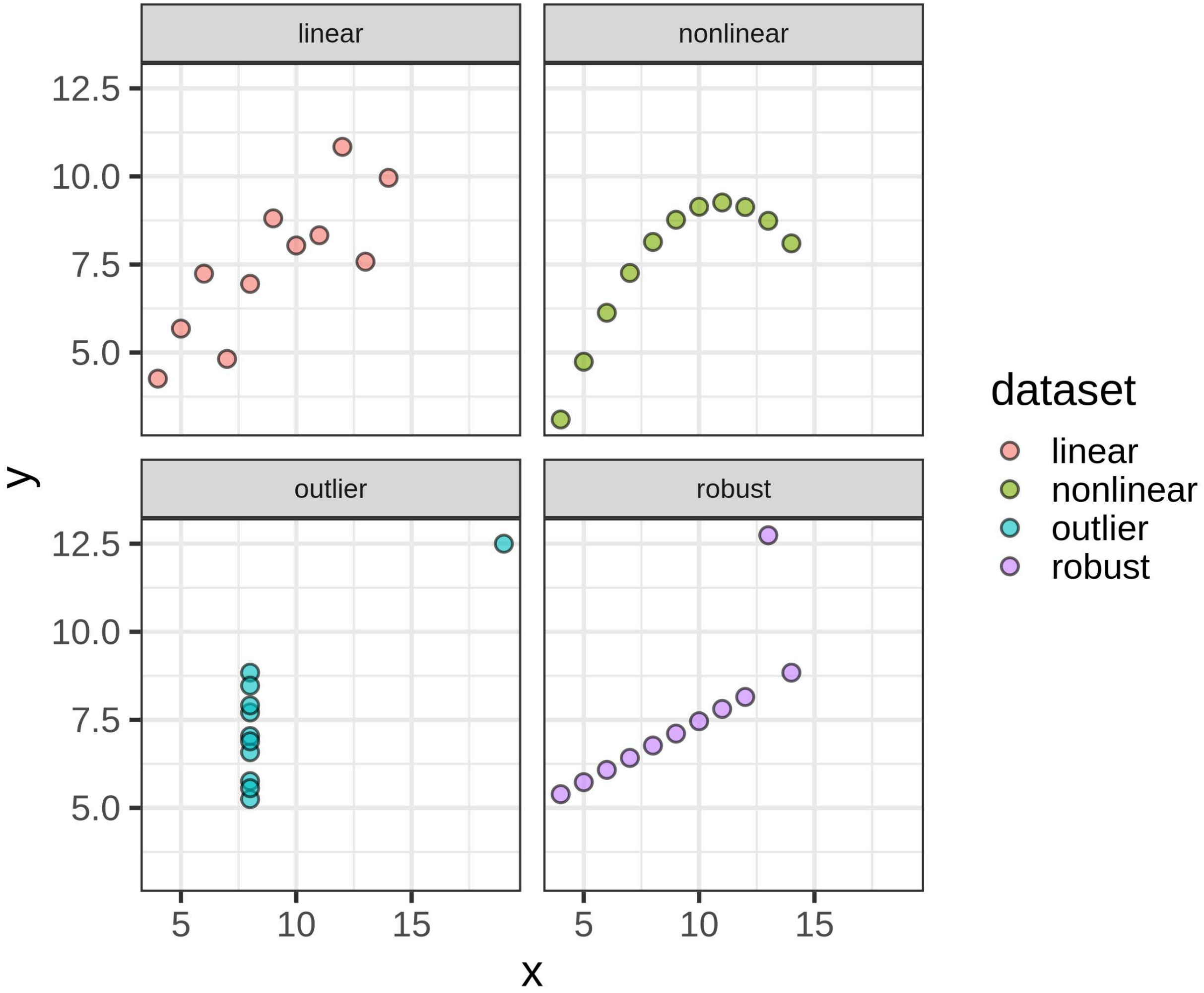
Correlation

- $-1 \leq r_{xy} \leq 1$
- no unit
- correlation of x with x equals 1
- unaffected by changes of units (e.g., cm to mm)
- can be interpreted without knowing the variances

Limitations

- Throughout, we're often looking at plots of the data
- The only thing that truly conveys all information in the data… are the data themselves
- Looking at raw numbers is typically not informative, hence we use plots

Anscombe's Quartet

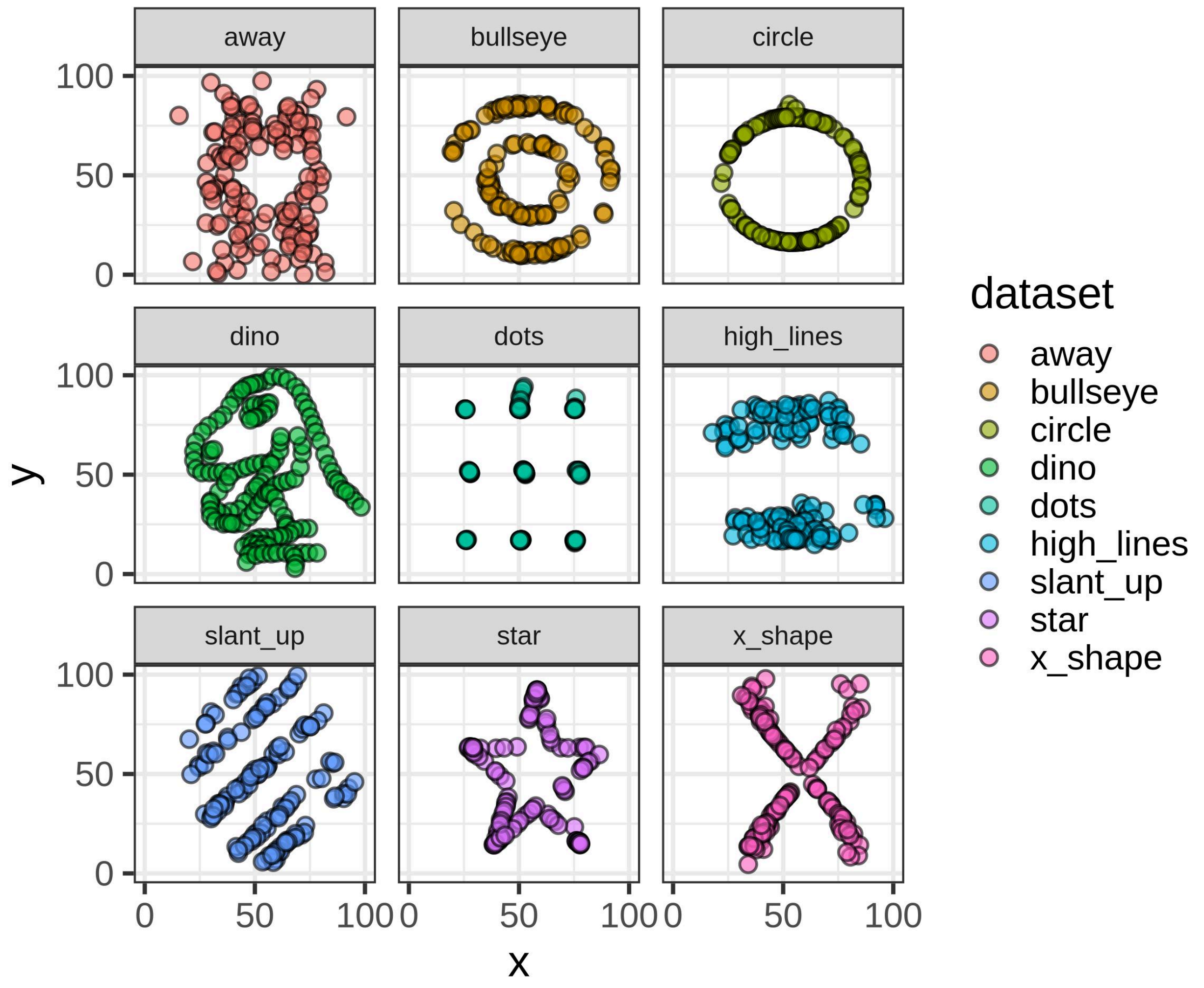


Source: Wikipedia

Anscombe’s Quartet

DATASET	MEAN_X	MEAN_Y	SD_X	SD_Y	COR
linear	9	7.501	3.317	2.032	0.816
nonlinear	9	7.501	3.317	2.032	0.816
outlier	9	7.501	3.317	2.031	0.817
robust	9	7.500	3.317	2.030	0.816

Datasaurus Dozen



source: https://jumpingrivers.github.io/datasauRus/

Datasaurus Dozen

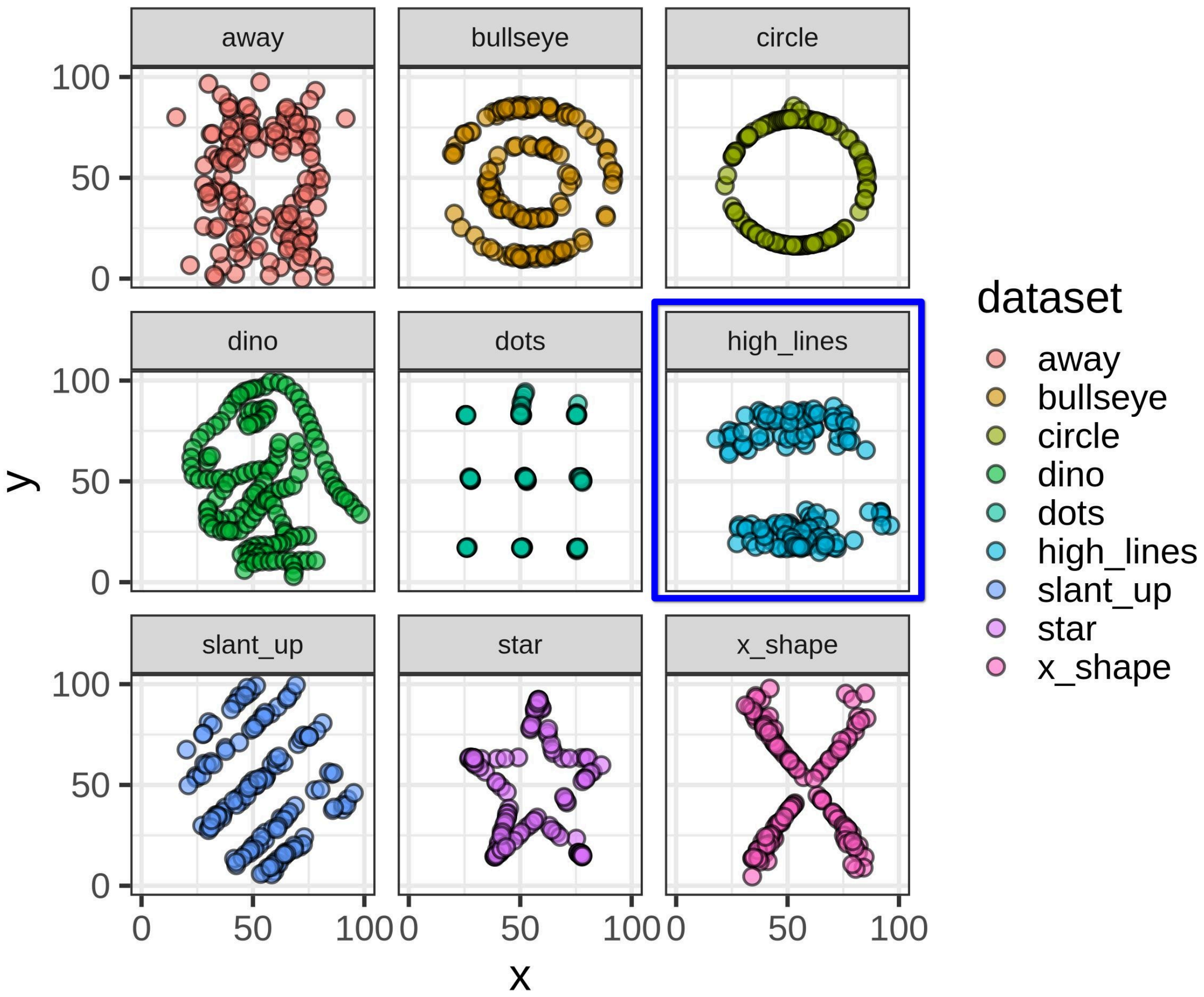
DATASET	MEAN_X	MEAN_Y	SD_X	SD_Y	COR
away	54.266	47.835	16.770	26.940	-0.064
bullseye	54.269	47.831	16.769	26.936	-0.069
circle	54.267	47.838	16.760	26.930	-0.068
dino	54.263	47.832	16.765	26.935	-0.064
dots	54.260	47.840	16.768	26.930	-0.060
high_lines	54.269	47.835	16.767	26.940	-0.069
slant_up	54.266	47.831	16.769	26.939	-0.069
star	54.267	47.840	16.769	26.930	-0.063

Limitations (2)

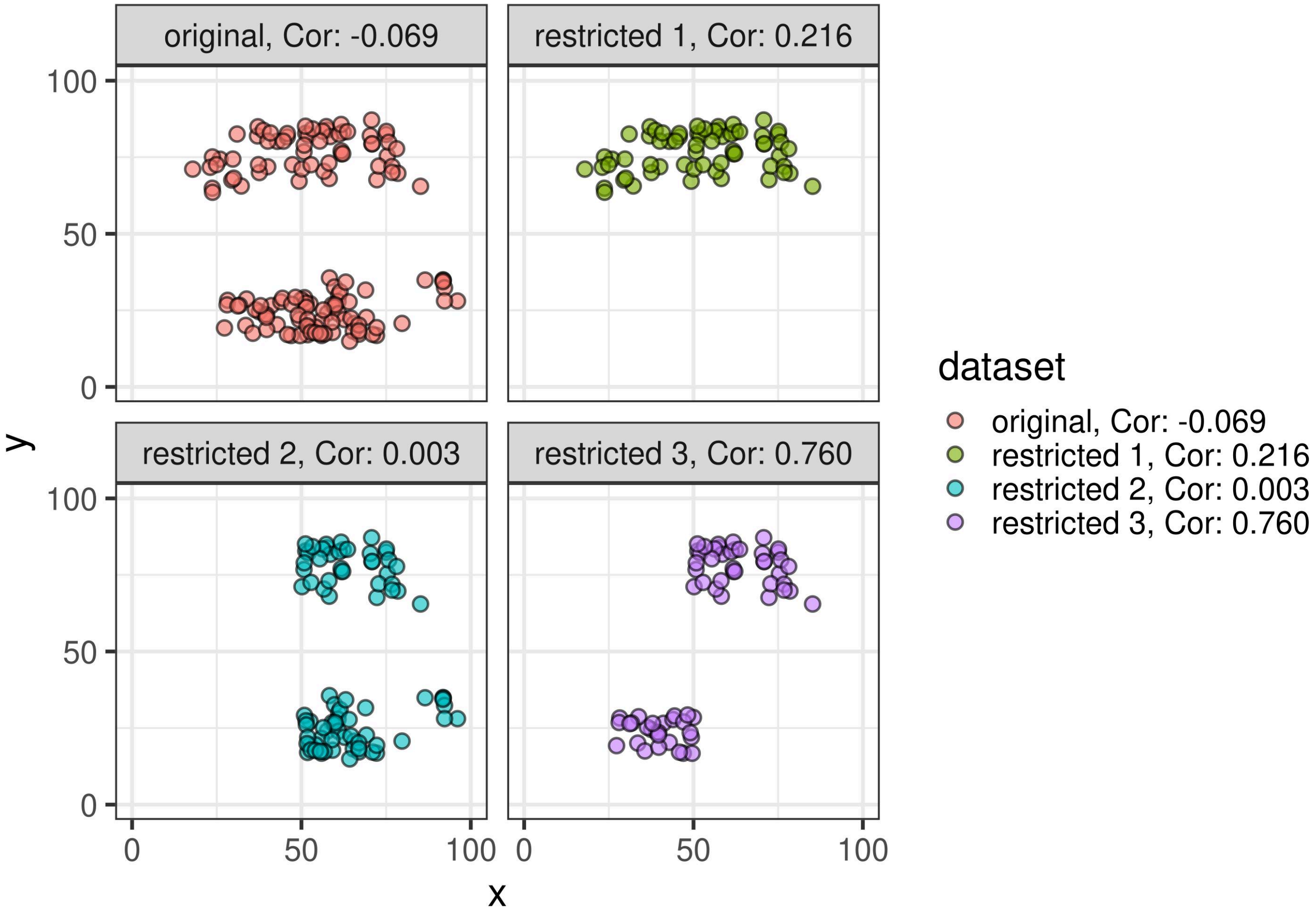
Some limitations cannot be spotted by looking at a plot!

1. Restricted range
2. Causality

Restricted range



Restricted range



Causality

What do these correlations tell us about causality, i.e., which variable *causes* a change in another?

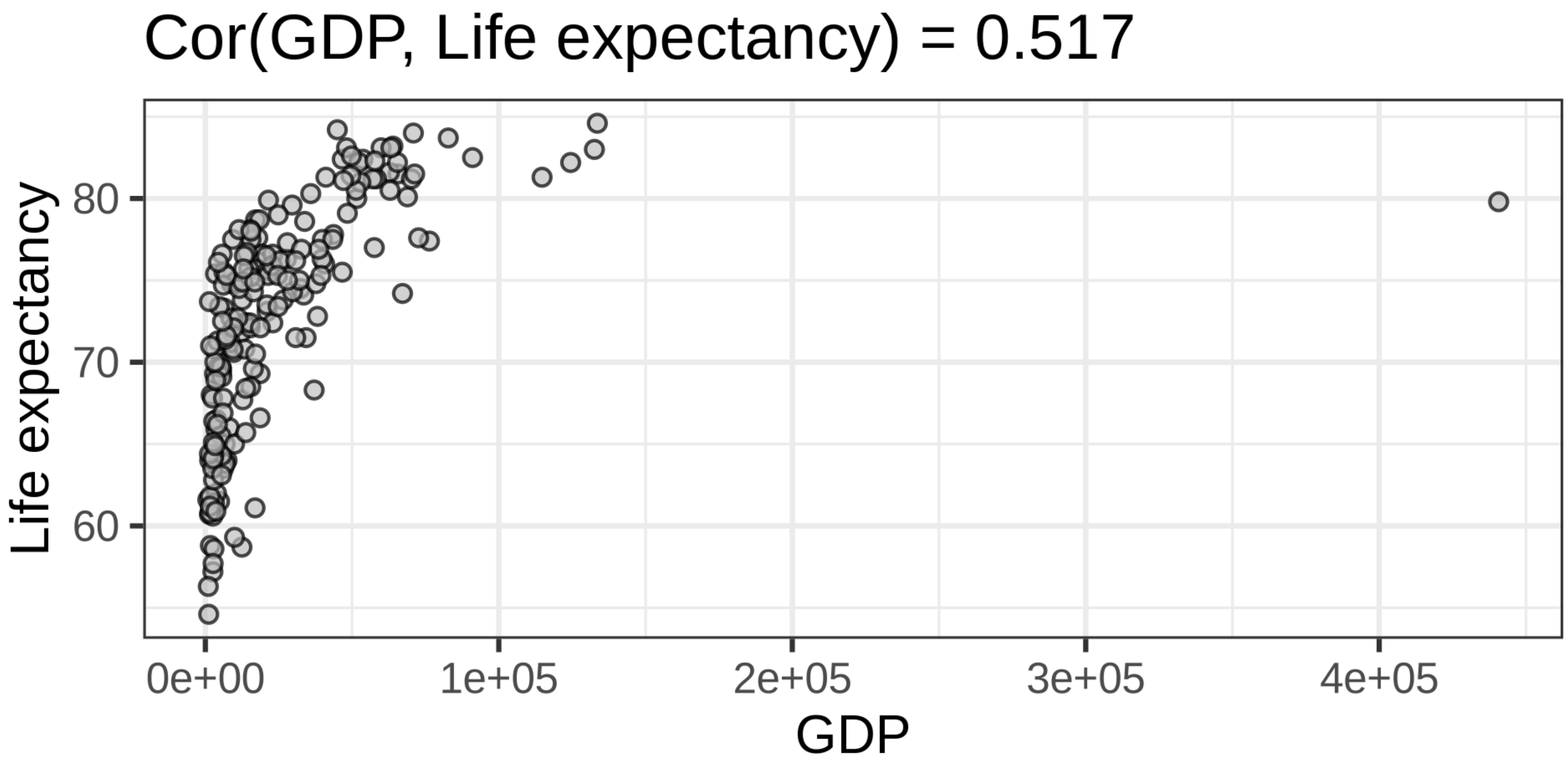
Nothing! *Correlation does not imply causation*!

Break

See you in ~10 minutes!

Transformations

Example: Wealth



What if the relation is not linear?

- Transformations
- Spearman rank correlation
- Kendall’s tau

Transformations

Transformations can make

- the data more normal
- relations more linear

which in turn makes statistics easier.

But, transformations also

- alter the meaning of the data
- are not guaranteed to make things better

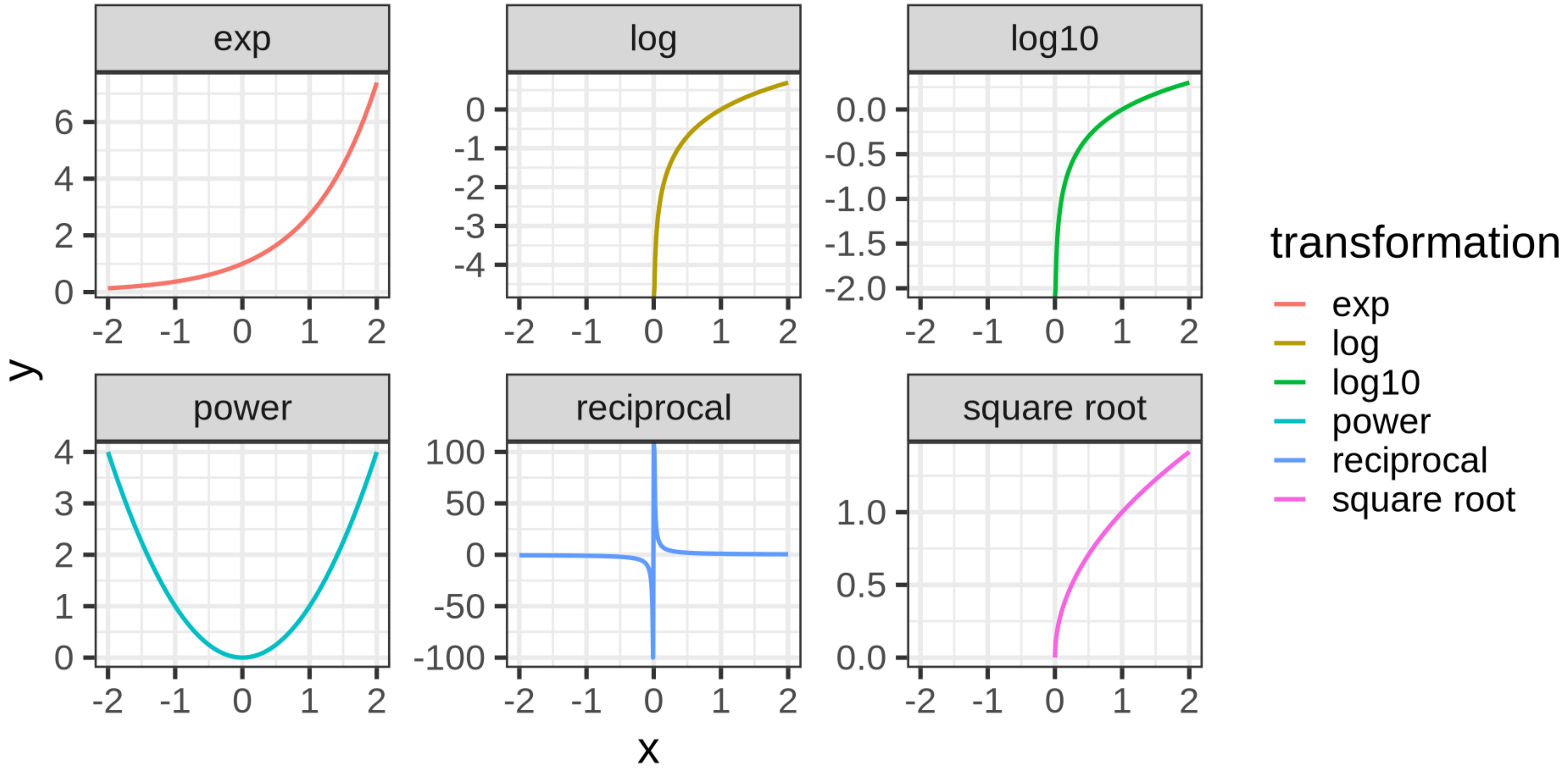
Common transformations

Some we have already seen!

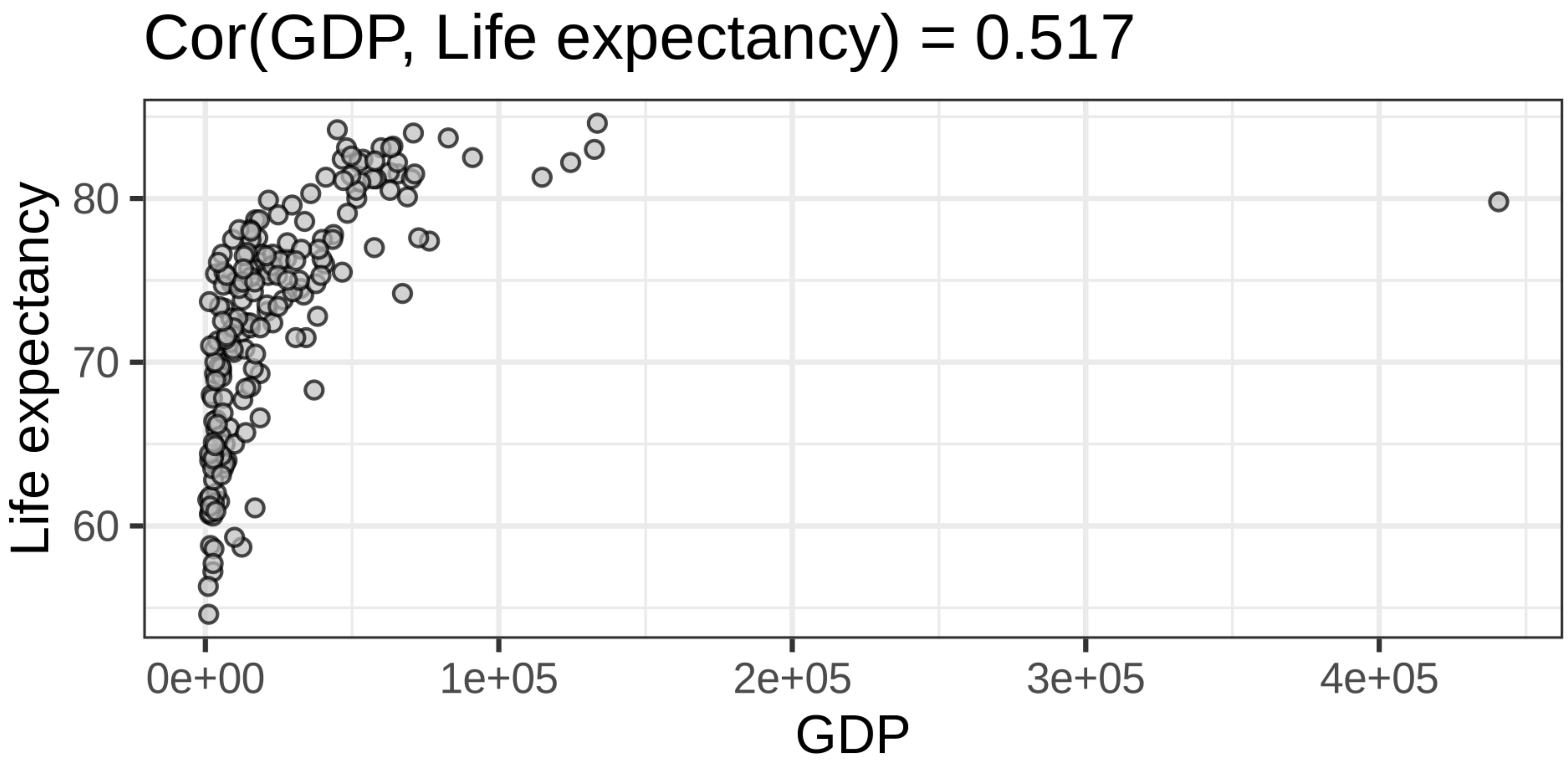
Converting raw to Z scores, is technically a transformation.

- Log, log10, reciprocal, exp, power, square root

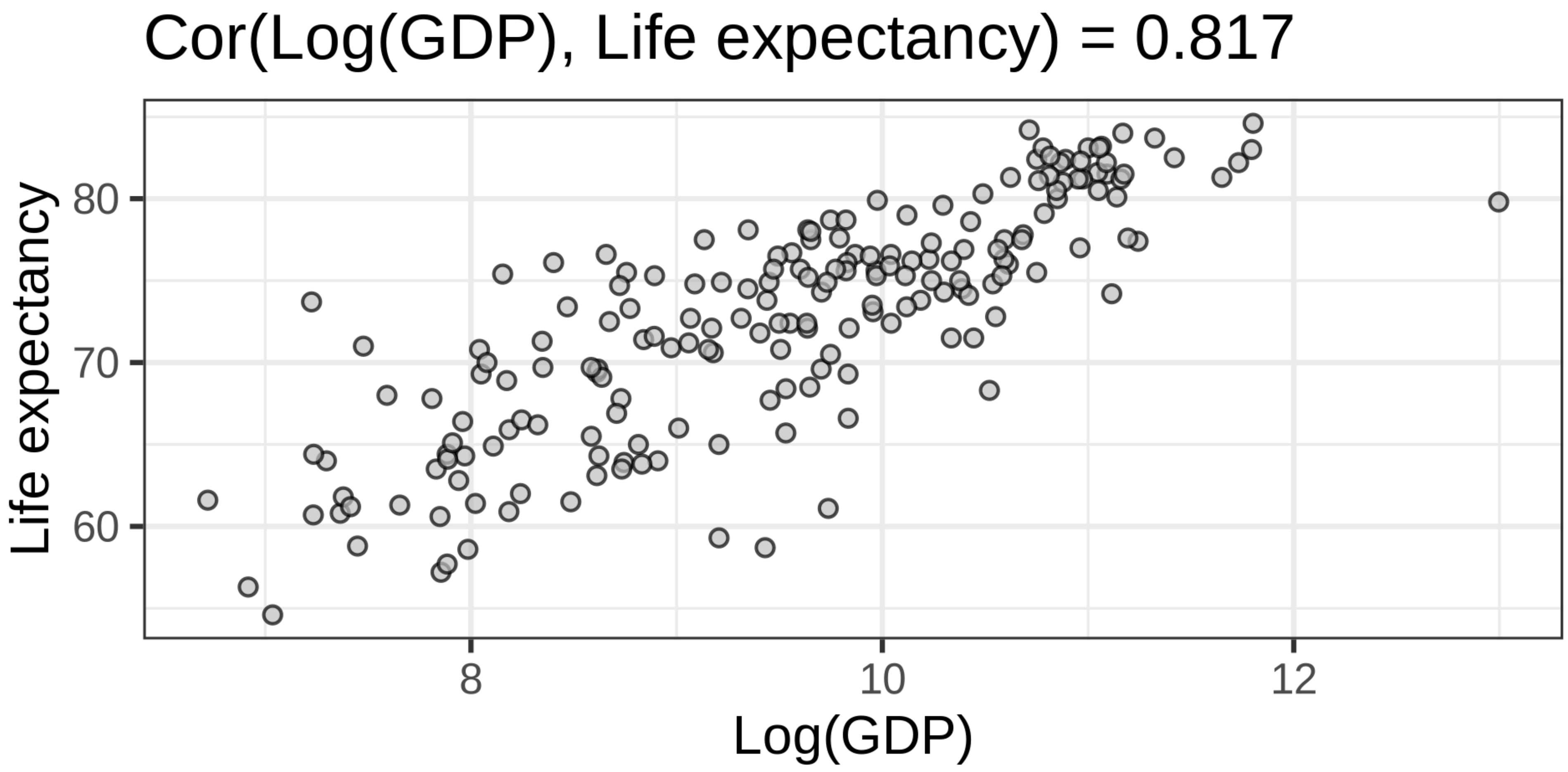
Common transformations



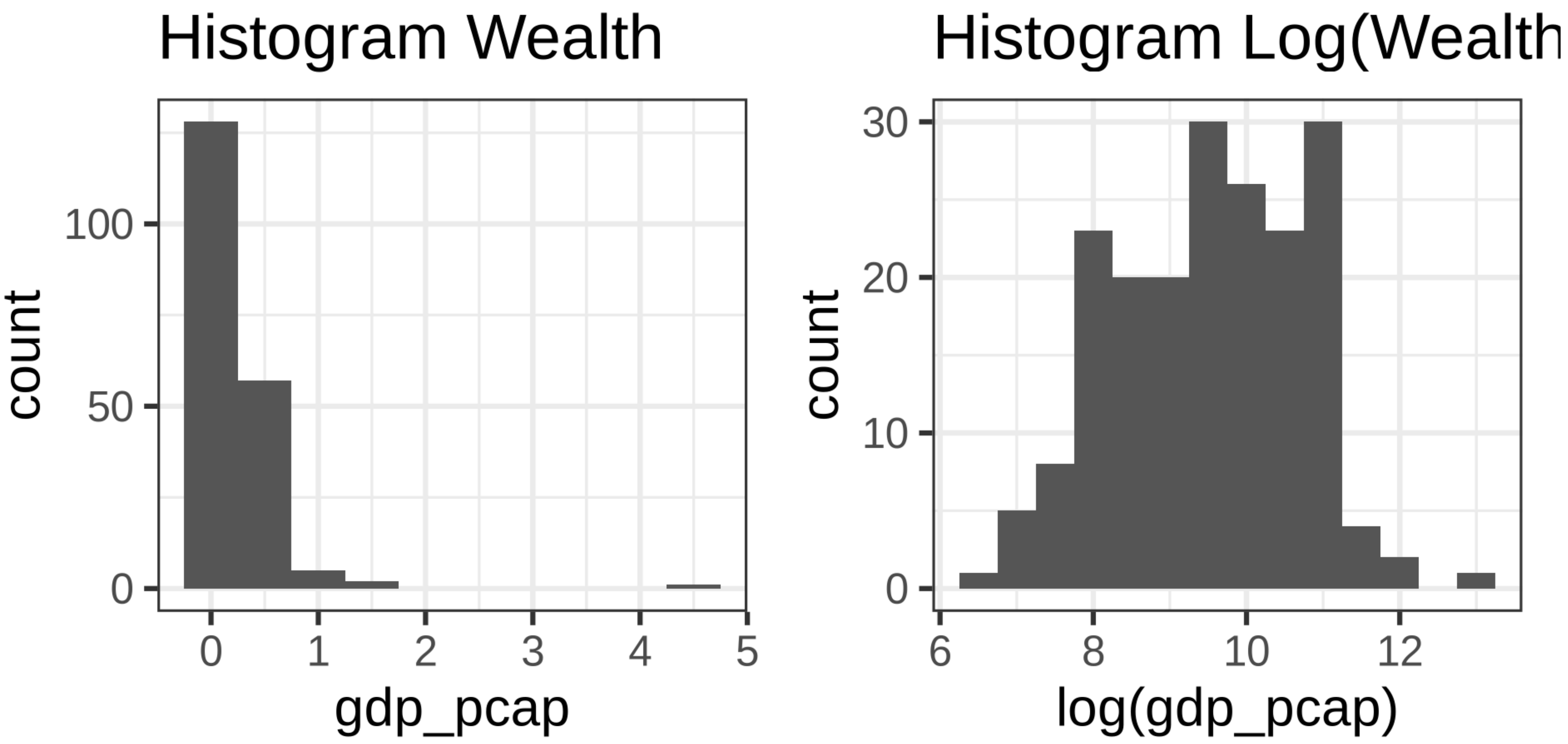
Example: Wealth



Example: Wealth



Example: Wealth



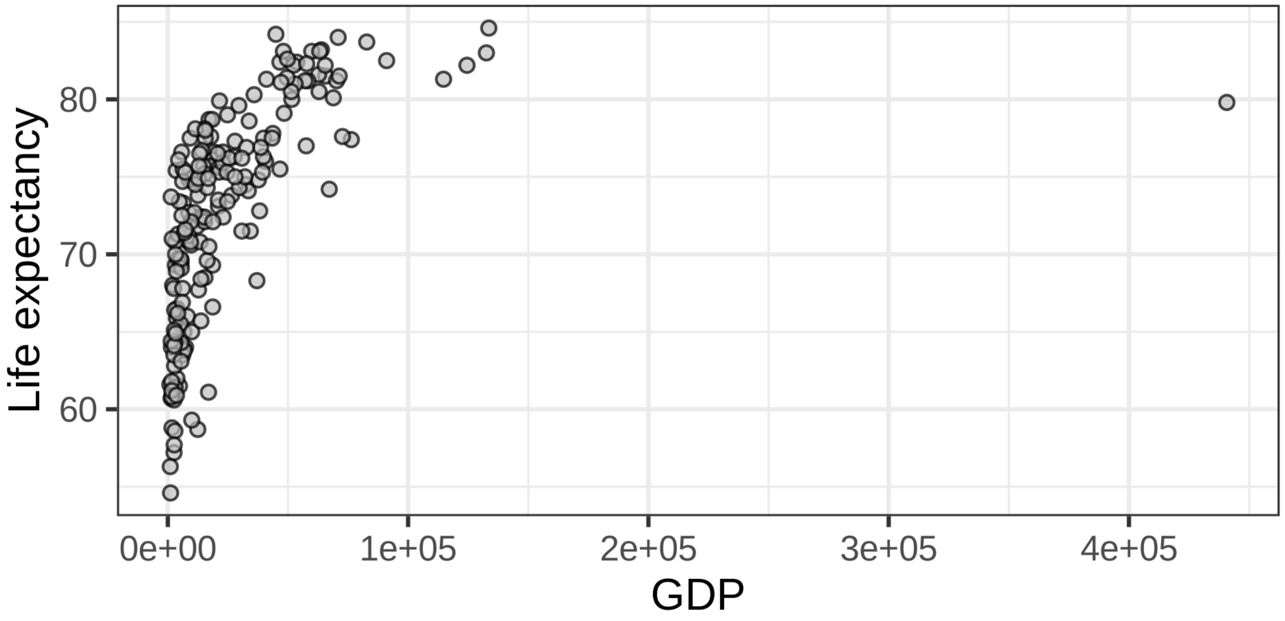
Spearman rank correlation

Pearson correlation between the rank values of those two variables

GDP_PCAP	RANK_GD	LIFE_EXPECTANCY	RANK_LIFE_EXPECTANCY
1,981.71	13.00	68.00	50.00
7,397.49	65.00	64.00	27.50
17,111.95	106.00	78.70	154.50
63,913.38	177.00	83.20	189.00
68,867.83	181.00	80.10	162.00
27,627.96	131.00	76.30	130.50

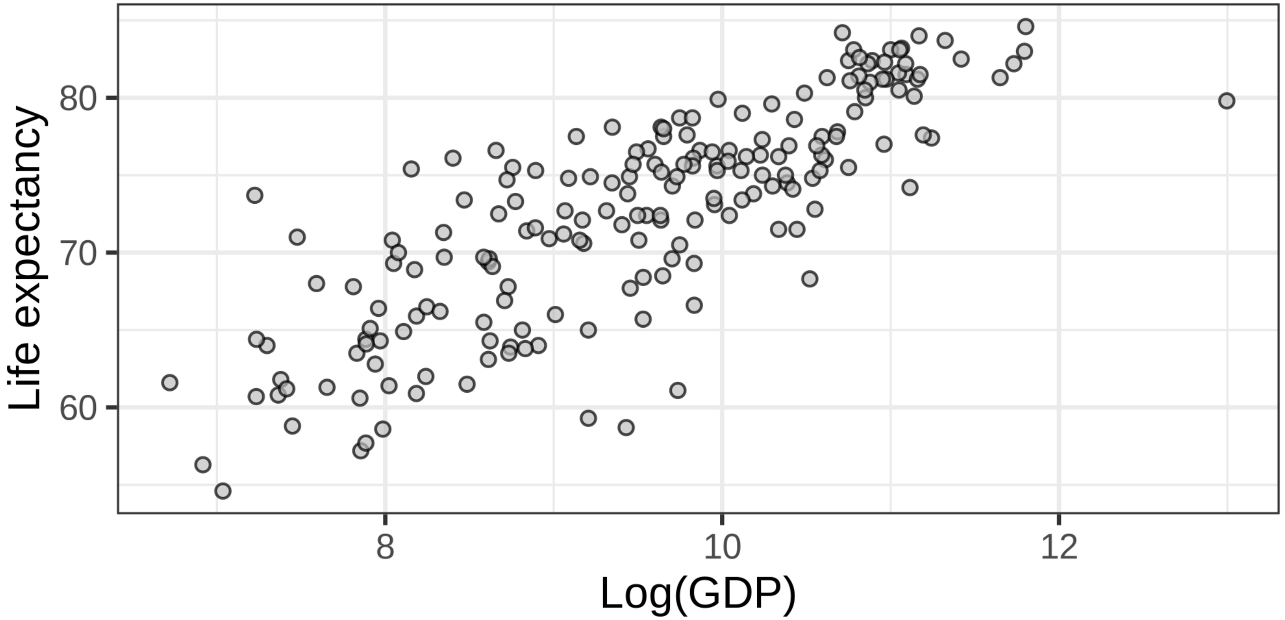
Example: Wealth

- Pearson Correlation: 0.517
- Spearman rank Correlation: 0.83
- Kendall's Tau: 0.638



Example: Wealth

- Pearson Correlation: 0.817
- Spearman rank Correlation: 0.83
- Kendall’s Tau: 0.638



Sample statistics and Population Estimators

- Sample statistics describe the sample.
- What we want to describe is the *population*.
- For this reason, sample statistics are often “adjusted”:

Sample

- Variance divides by n
- Covariance divides by n

Estimator

- Variance divides by $n - 1$
- Covariance divides by $n - 1$

Intuition: We “used” one observation to compute the sample mean and thus divide by $n - 1$.

Note:

- Variance and covariance change when we use $n - 1$ to estimate population quantities
- Correlation does not change, because the denominator cancels.

Drawbacks of Descriptive Statistics

At its core, statistics tries to *reduce* the information in the full dataset into something smaller

- (+) This makes it possible to understand large datasets
- (-) This loses information along the way

Example: **IF** the data are normally distributed then the mean and variance describe the entire distribution.

Bivariate Distributions

Normal distribution

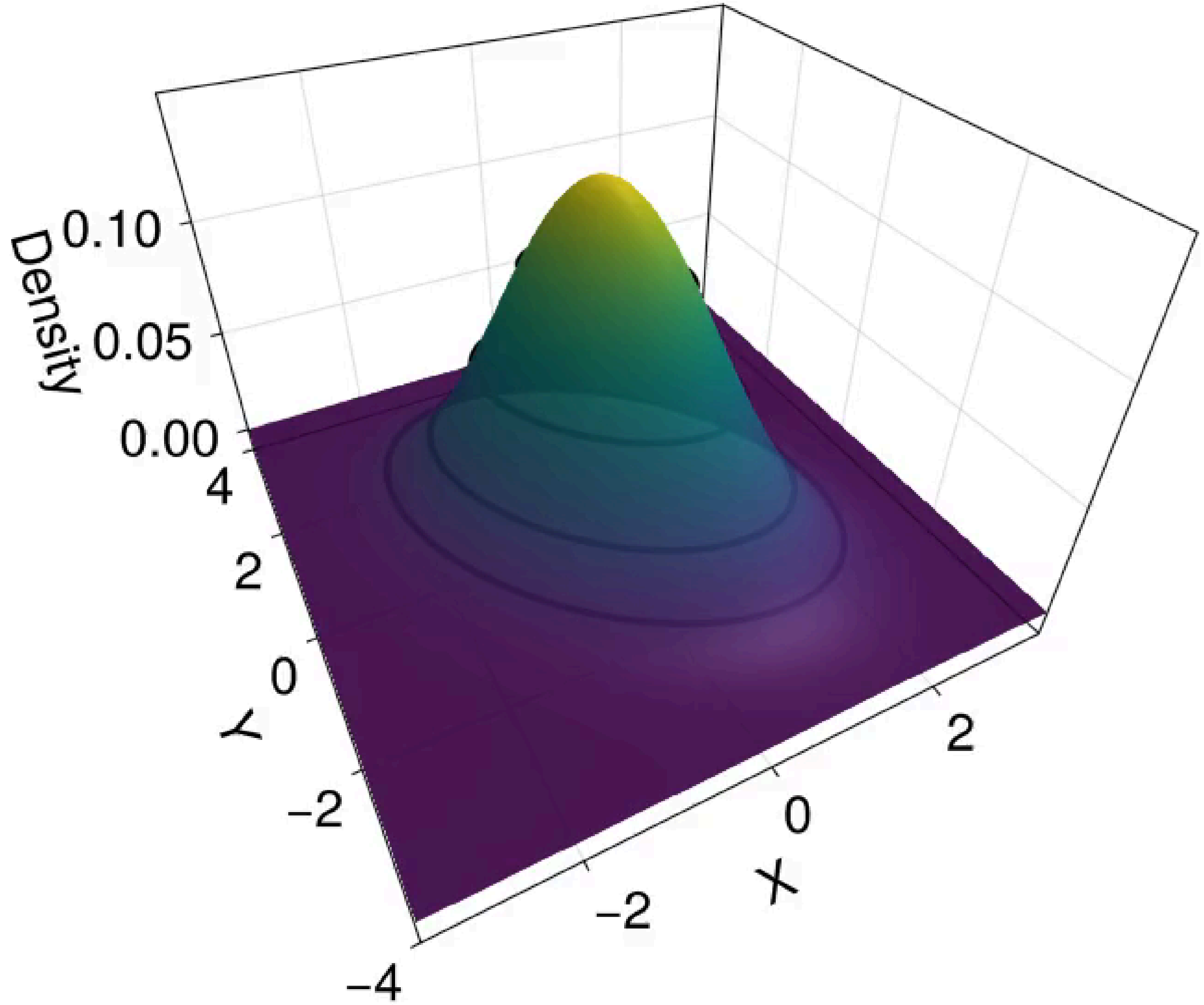
Last Week: Sample mean and SD correspond to the population model $N(\mu_x, \sigma_x)$

Now: two sample means, $\bar{x}$, $\bar{y}$, two SDs, s_x, s_y, a correlation r_{xy} correspond to a bivariate normal model:

$$X, Y \sim N(\underbrace{\mu_x, \mu_y}_{\text{mean}}, \underbrace{\sigma_x, \sigma_y, \rho}_{\text{covariance}})$$

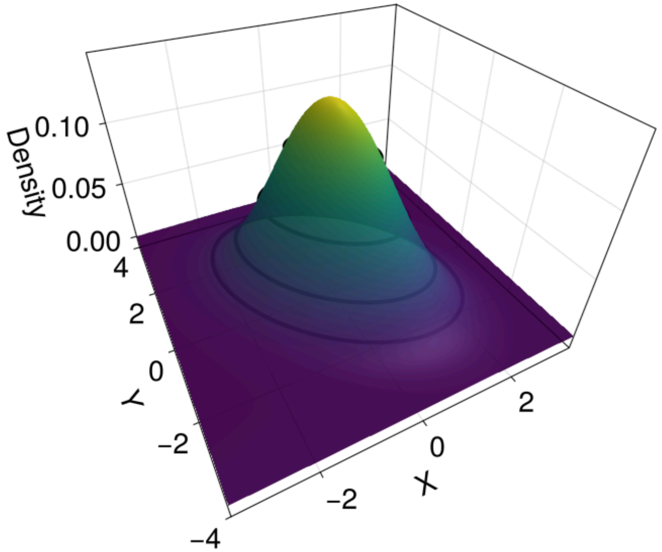
The bivariate normal distribution has five parameters:

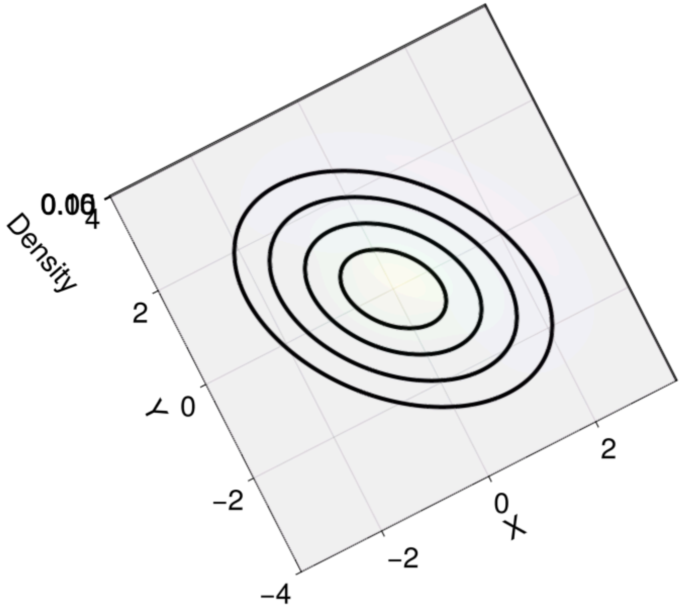
$$\begin{aligned} \mu_x &: \text{mean of X} & \sigma_x &: \text{sd of X} \\ \mu_y &: \text{mean of Y} & \sigma_y &: \text{sd of Y} \\ & & \rho &: \text{correlation} \end{aligned}$$



Density
0.10
0.05
0.00
4
2
0
Y
−2
−4
−2
0
2
X

These are the same picture





Bivariate Normal distribution

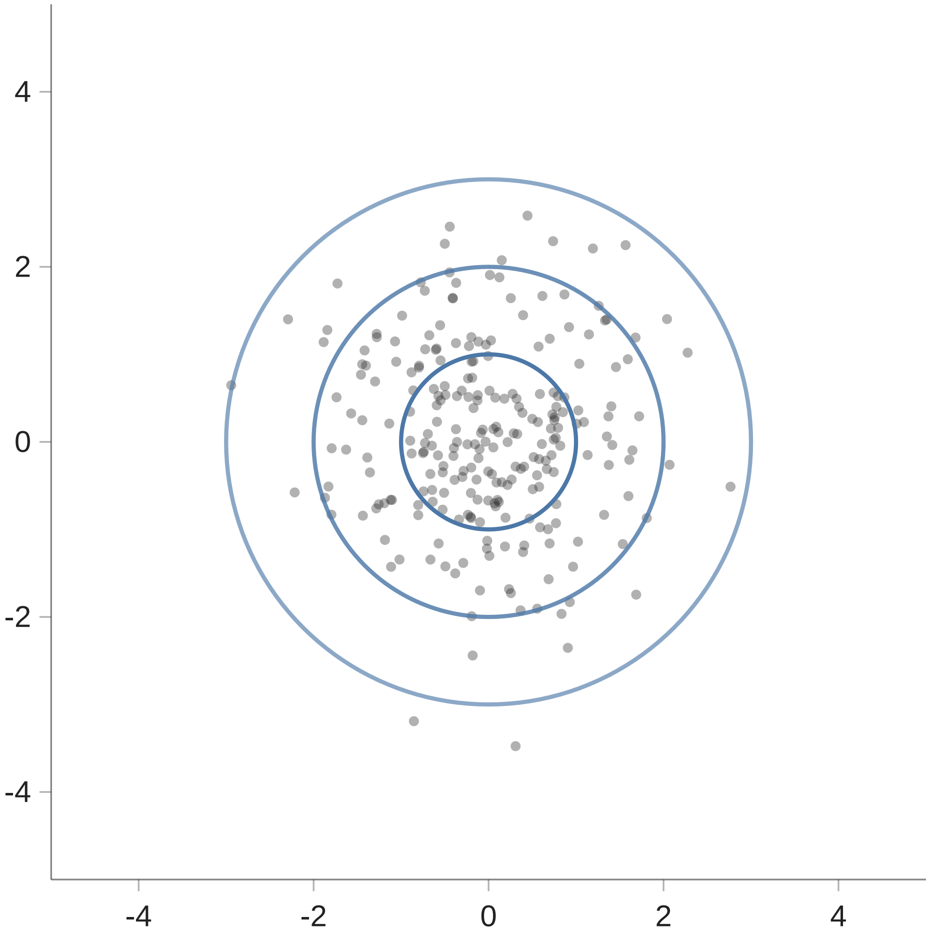
μ_x 0.0

μ_y 0.0

σ_x 1.0

σ_y 1.0

ρ 0.0



Continuous Distributions

Last week we examined discrete probability distributions:

- joint, marginal, and conditional.

For continuous probability distributions the exact counterparts exist.

Marginal Distribution

If we ignore Y, or have not observed it, what is the distribution of X?

Conveniently, if

$$X, Y \sim N(\underbrace{\mu_x, \mu_y}_{\text{mean}}, \underbrace{\sigma_x, \sigma_y, \rho}_{\text{(Co)Variance}})$$

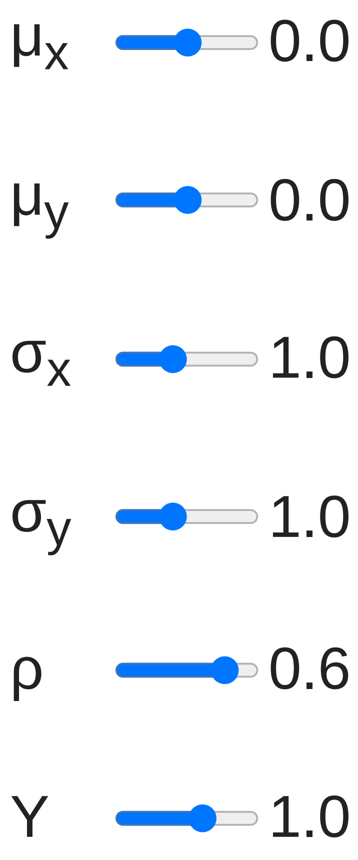
then the marginal distributions are

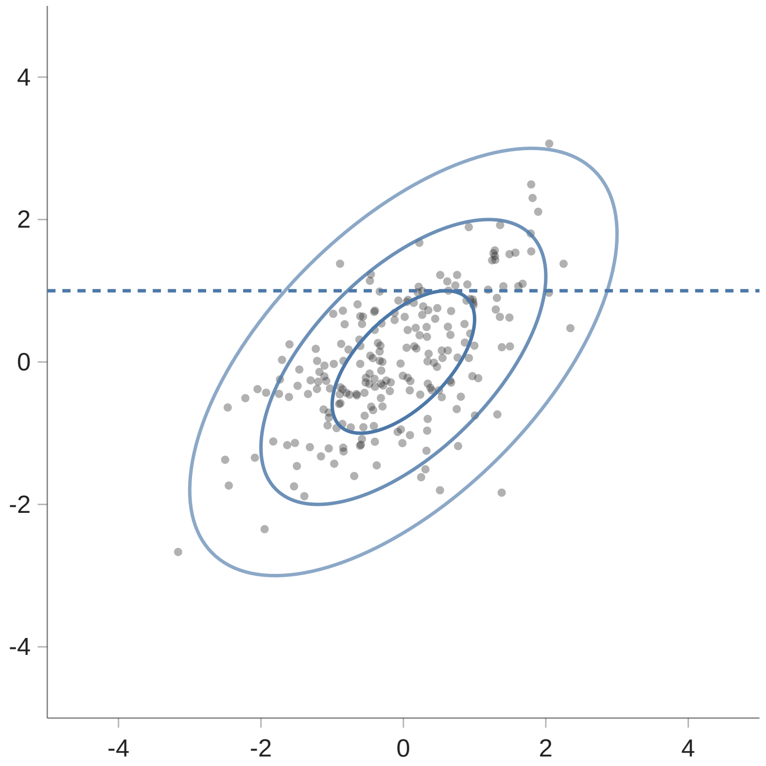
$$X \sim N(\mu_x, \sigma_x), \quad Y \sim N(\mu_y, \sigma_y)$$

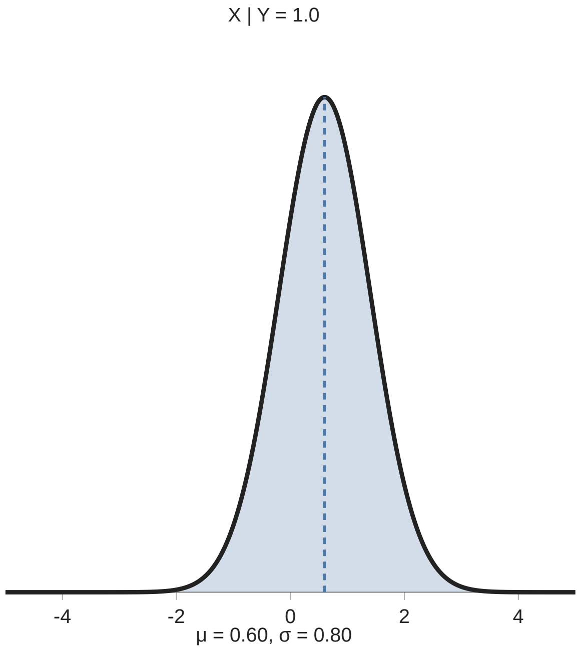
Conditional Distribution

What if we do know Y?

Conditional Distribution







Conditional Distribution

If $\rho = 0$ then knowing Y does not tell us anything about X.

They are *independent*.

Independence: knowing the outcome or value of one variable or event provides no information about the probability or outcome of another.

This is a special property of the bivariate normal distribution, it does *not* hold in general that $\rho = 0$ (or $r = 0$) implies independence.

Multivariate Normal distribution

For p variables we have:

$$\mu = \begin{bmatrix} \mu_1 \\ \mu_2 \\ \mu_3 \\ \vdots \\ \mu_p \end{bmatrix} \qquad \mathbf{\Sigma} = \begin{bmatrix} \sigma_1^2 & \sigma_1\sigma_2\rho_{12} & \sigma_1\sigma_3\rho_{13} & \cdots & \sigma_1\sigma_p\rho_{1p} \\ \sigma_2\sigma_1\rho_{21} & \sigma_2^2 & \sigma_2\sigma_3\rho_{23} & \cdots & \sigma_2\sigma_p\rho_{2p} \\ \sigma_3\sigma_1\rho_{31} & \sigma_3\sigma_2\rho_{32} & \sigma_3^2 & \cdots & \sigma_3\sigma_p\rho_{3p} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \sigma_p\sigma_1\rho_{p1} & \sigma_p\sigma_2\rho_{p2} & \sigma_p\sigma_3\rho_{p3} & \cdots & \sigma_p^2 \end{bmatrix}$$

- No more picture
- If $\rho_{ij} = 0$ then variables i and j are independent

Summary

- Covariance and correlation summarize the association between two variables.
- Transformations and rank correlations may help when the relation is nonlinear.
- The bivariate normal distribution describes the joint, marginal, conditional distributions and independence.